6.Create dashboards in Tableau Public Edition using a COVID-19 (compact) dataset:

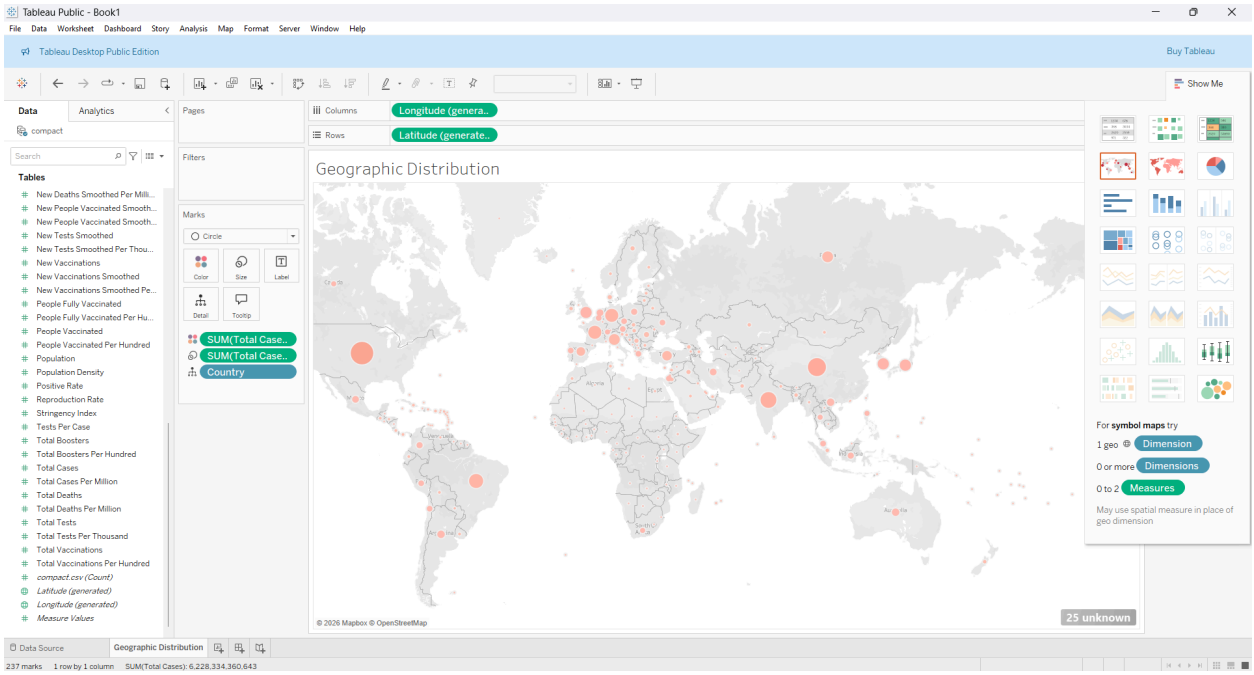
a. Global COVID-19 Tracker Dashboard: - Create a dashboard to track global COVID-19 cases, deaths, and recoveries
i. Map to show geographic distribution of cases
ii. Bar chart to show top 10 countries with most cases
iii. Line chart to show daily new cases and deaths Add filters for date range and country/region

Step 1: Connect to the Data

1. Open **Tableau Public**.
2. In the **Connect** pane on the left, click **Text File**.
3. Select your `compact1.csv` file.

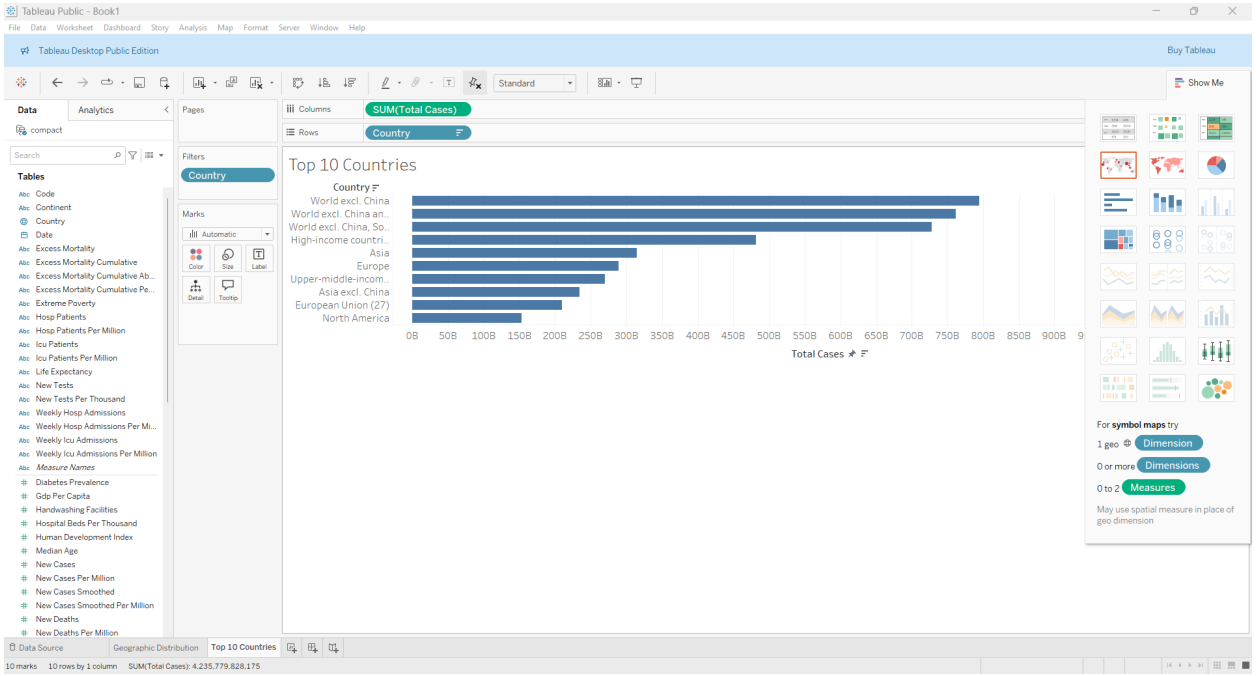
Step 2: Create Sheet 1 – Global Distribution Map

1. Double-click the **Country/Region** dimension. Tableau will automatically generate a map.
2. Drag the **Confirmed** (or Cases) measure to the **Color** mark.
3. Drag **Confirmed** again to the **Size** mark (optional, to make dots larger for higher cases).
4. Click on **Color** in the Marks card and select "Edit Colors" (e.g., Red-Gold) to make it look like a heat map.
5. Rename the sheet: `Geographic Distribution`.



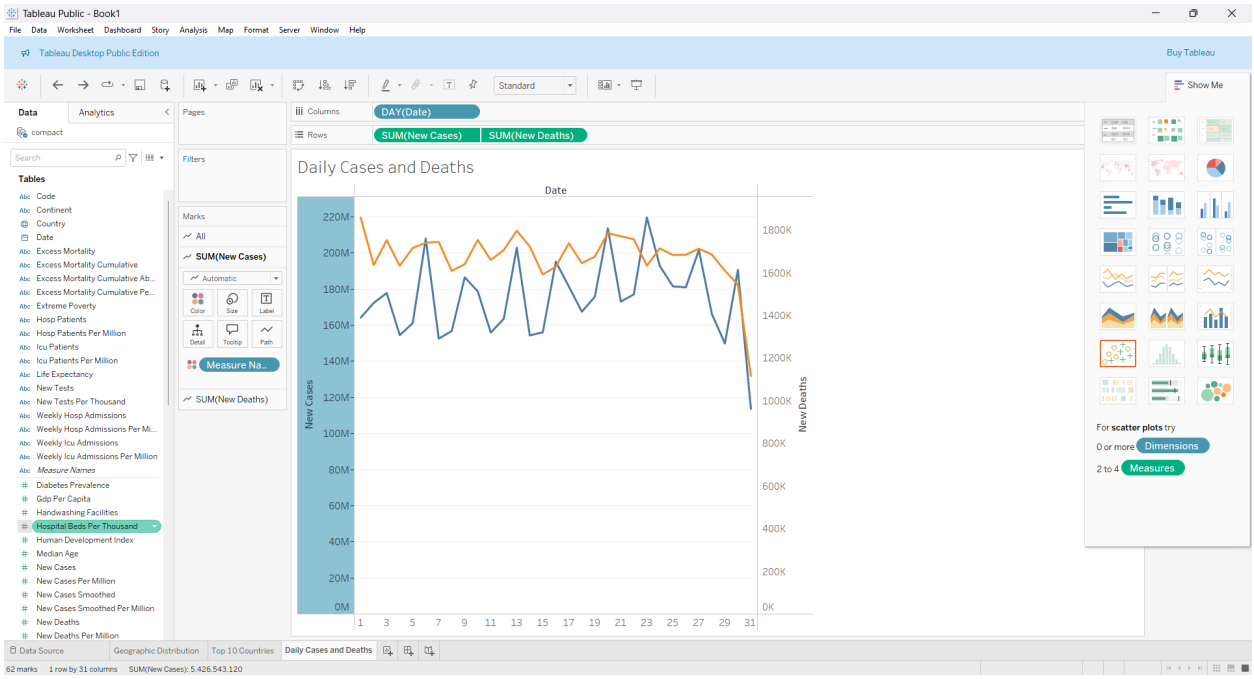
Step 3: Create Sheet 2 – Top 10 Countries (Bar Chart)

1. Click the "New Worksheet" icon.
2. Drag **Country/Region** to the **Rows** shelf.
3. Drag **Confirmed** to the **Columns** shelf.
4. Click the **Sort Descending** icon on the toolbar.
5. **To filter for Top 10:**
    - Drag **Country/Region** into the **Filters** shelf.
    - Go to the **Top** tab.
    - Select **By field** -> **Top** -> 10 -> by **Confirmed** (Sum).
    - Click **OK**.
6. Rename the sheet: Top 10 Countries.



Step 4: Create Sheet 3 – Daily Trend (Line Chart)

1. Click the "New Worksheet" icon.
2. Drag **Date** to the **Columns** shelf.
    - *Tip: Click the '+' on the Year pill or right-click it and select 'Day' or 'Month' to change the granularity.*
3. Drag **Confirmed** (New Cases) and **Deaths** to the **Rows** shelf.
4. To put them on the same graph: Right-click the **Deaths** pill on the Rows shelf and select **Dual Axis**.
5. Change the Marks type to **Line**.
6. Rename the sheet: `Daily Cases and Deaths`.



Step 5: Build the Dashboard and Add Filters

1. Click the **New Dashboard** icon (at the bottom).
2. On Left Side Change **Size** drop down to **Automatic**
3. **Drag and Drop:** Pull all three sheets onto the dashboard canvas.
4. **Add Global Filters:**
    - Go back to Sheet 1. Drag **Date** and **Country/Region** to the **Filters** shelf.
    - Right-click each filter in the shelf -> **Apply to Worksheets** -> **All Using This Data Source**.
    - This ensures that when you filter for a country on the map, the Bar and Line charts update instantly.
5. **Show Filters on Dashboard:**
    - Click on one of the sheets in the dashboard.
    - Click the small down arrow in the top-right corner of the sheet -> **Filters** -> select **Date** and **Country/Region**.

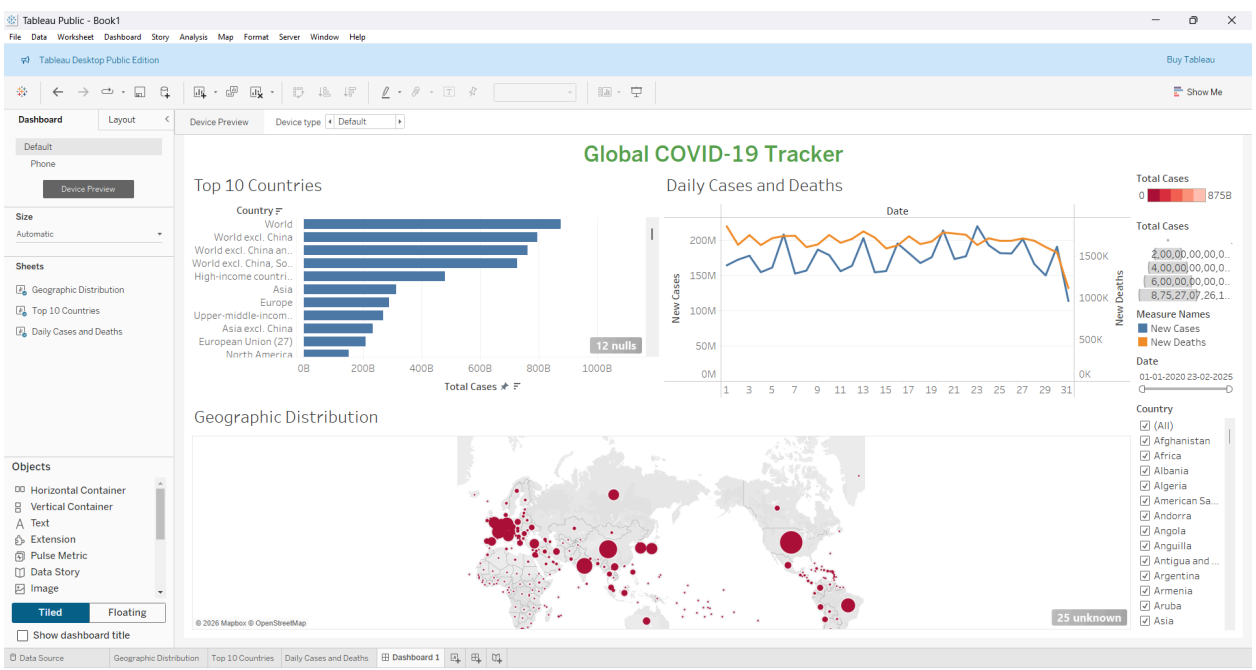
<u>Adding a Dashboard Title</u>

In the **Objects** pane (bottom-left), find the **Text** icon.

Drag and Drop the Text object onto the dashboard where you want the label.

An "Edit Text" window will pop up. Type your title, select the font size (e.g., 20pt), make it **Bold**, and click **OK**.

To move it: Click the text box and drag it by the "grip" (the dark gray bar at the top of the box).

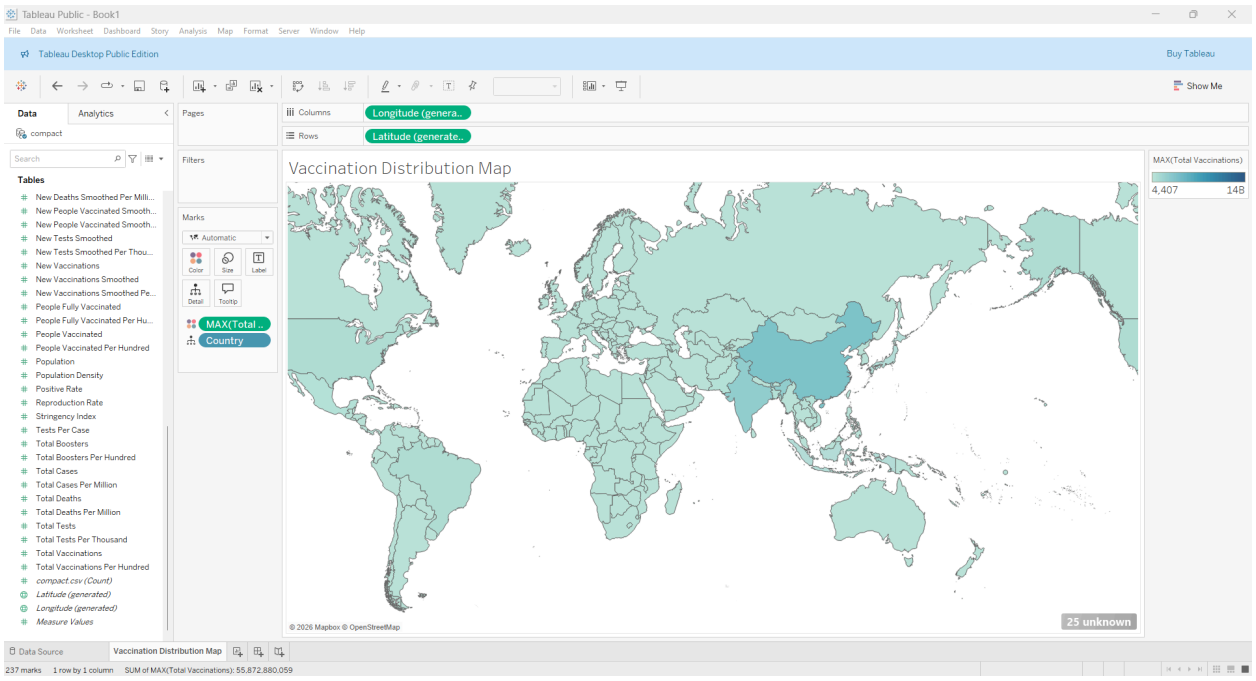


b. COVID-19 Vaccination Progress Dashboard
i. Map to show geographic distribution of vaccination rates
ii. Bar chart to show top 10 countries with highest vaccination rates
iii. Line chart to show daily new vaccinations Add filters for date range and country/region

Step 1: Create Sheet 1 – Vaccination Map

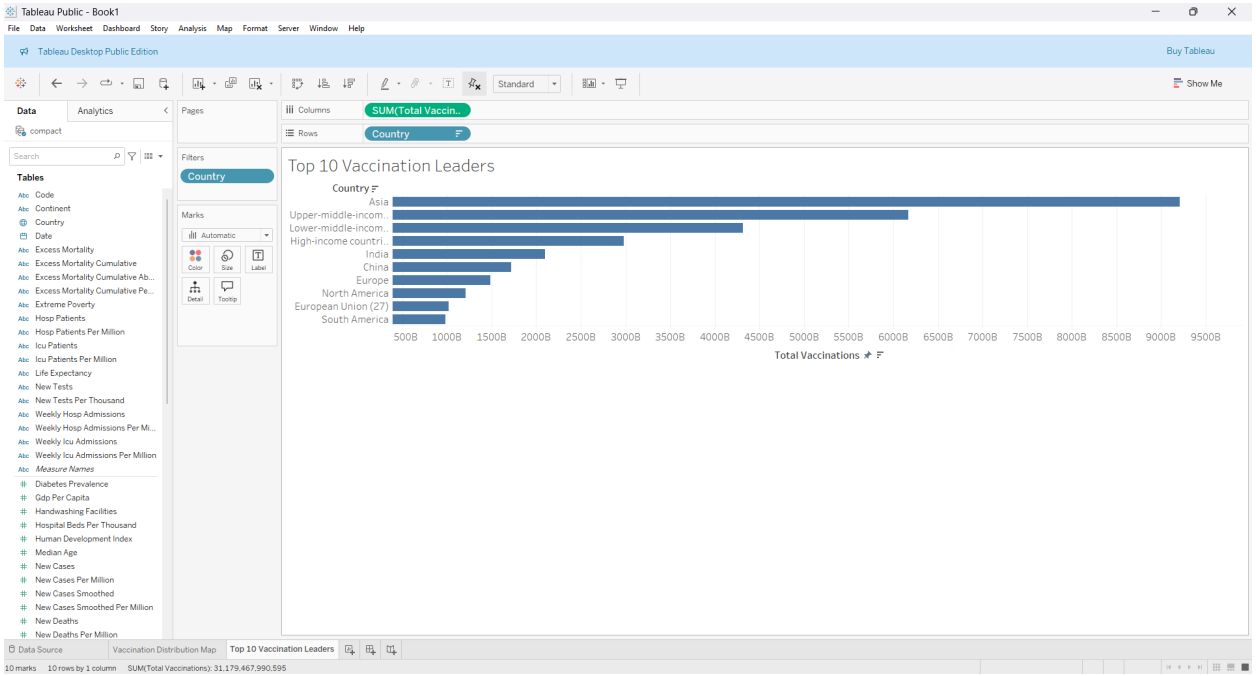
This shows which countries are leading in their rollout.

1. Click the New **Worksheet** icon.
2. Double-click **Country/Region**. Tableau generates the map.
3. Drag **Total Vaccinations** (or *Vaccination Rate* if your CSV has it) to Color.
4. If your data has "Total Vaccinations" as a cumulative number, right-click the green pill in the Marks card, go to **Measure**, and ensure it is set to **Maximum** (to show the most recent progress) rather than Sum.
5. Rename the sheet: `Vaccination Distribution Map`.



Step 2: Create Sheet 2 – Top 10 Countries (Highest Rates)

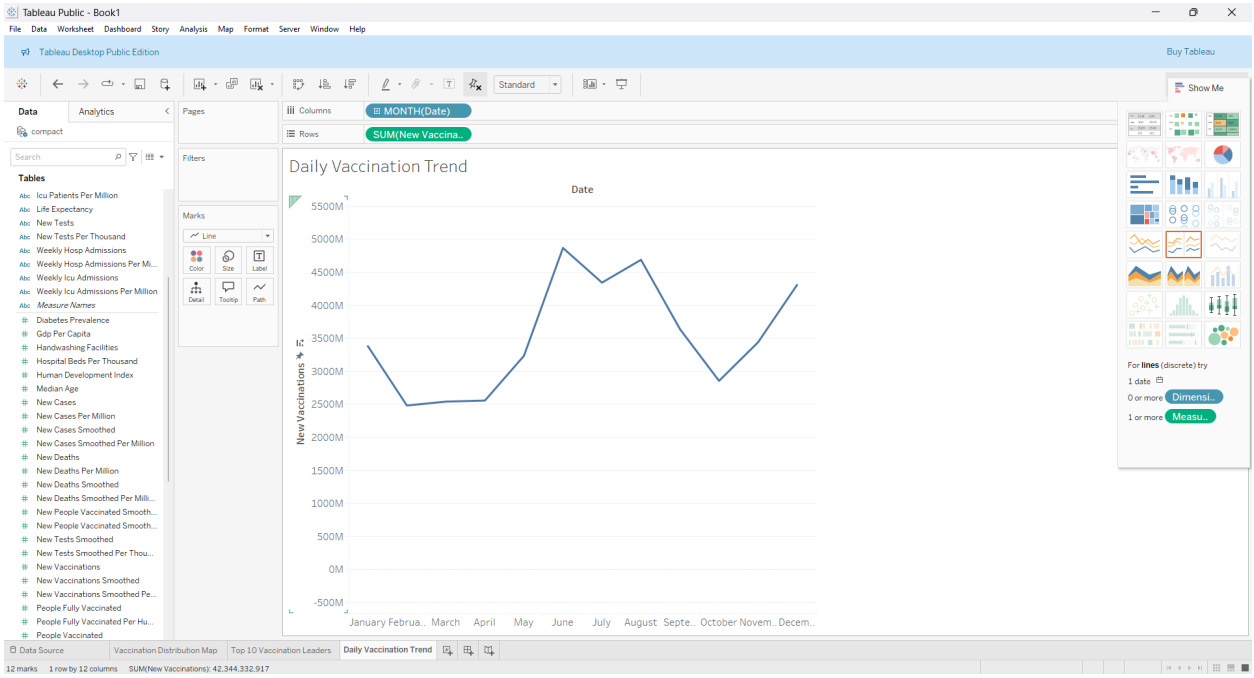
1. Click the **New Worksheet** icon.
2. Drag **Country/Region** to **Rows**.
3. Drag **Total Vaccinations** to **Columns**.
4. **Filter for Top 10:**
    - Drag **Country/Region** to the **Filters** shelf.
    - Go to the **Top** tab -> **By field** -> **Top 10** -> by **Total Vaccinations** (Maximum).
5. Sort the bars in descending order using the icon in the top toolbar.
6. Rename the sheet: Top 10 Vaccination Leaders.



Step 3: Create Sheet 3 – Daily Vaccination Trend

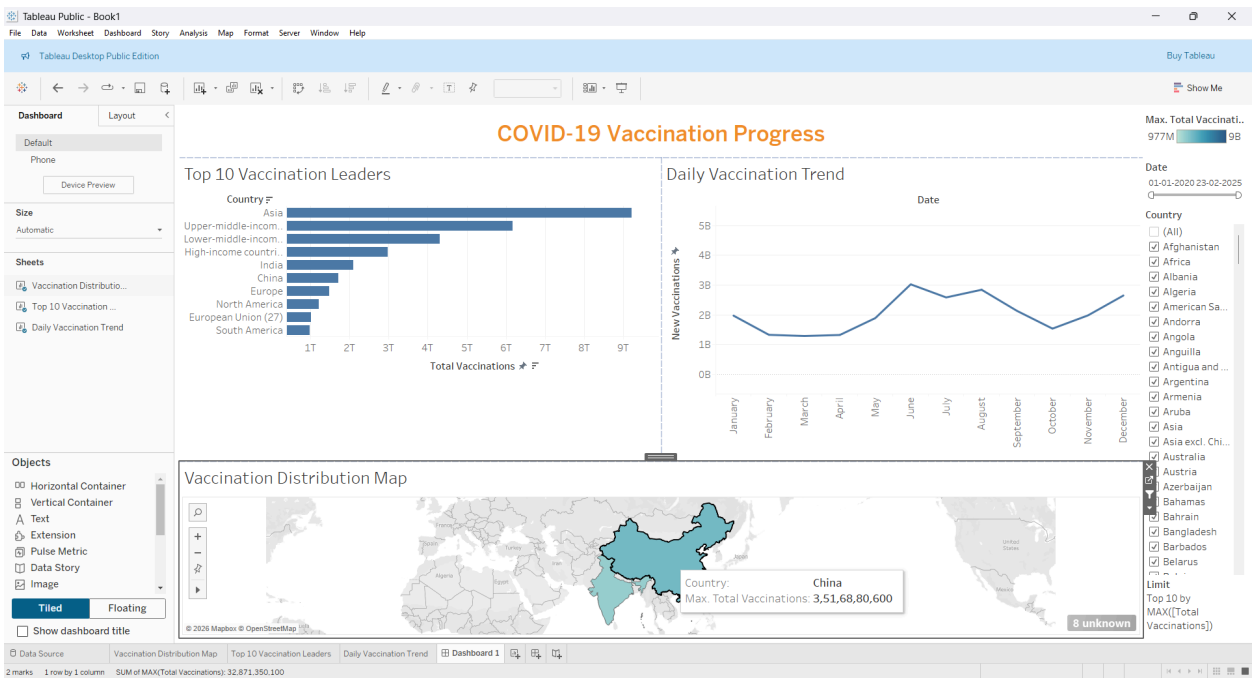
This tracks the speed of the rollout over time.

1. Click the **New Worksheet** icon.
2. Drag **Date** to the **Columns** shelf (Change to **Day** or **Month** granularity).
3. Drag **New Vaccinations** to the **Rows** shelf.
    - *Note: If your dataset only has "Total Vaccinations," you can right-click the pill -> Quick Table Calculation -> Difference, to see the daily change.*
4. Change the Mark type to **Line** or **Area Chart**.
5. Rename the sheet: Daily Vaccination Trend.



Step 4: Build the Dashboard

1. Click **New Dashboard**.
2. Set the Size to **Automatic** so it fills your screen.
3. Drag all three Vaccination sheets onto the canvas.
4. **Add Labels:**
    - Drag a **Text Object** to the top and type: COVID-19 Vaccination Progress Dashboard.
5. **Sync the Filters:**
    - Go to any sheet, right-click the **Date** filter -> **Apply to Worksheets** -> **All using this data source**.
    - Do the same for the **Country** filter.
6. **Interactive Map:**
    - Click on the Map on the dashboard and click the **Use as Filter** (funnel icon). Now, clicking "India" or "USA" on the map will instantly update the trend line and bar chart for that specific country.

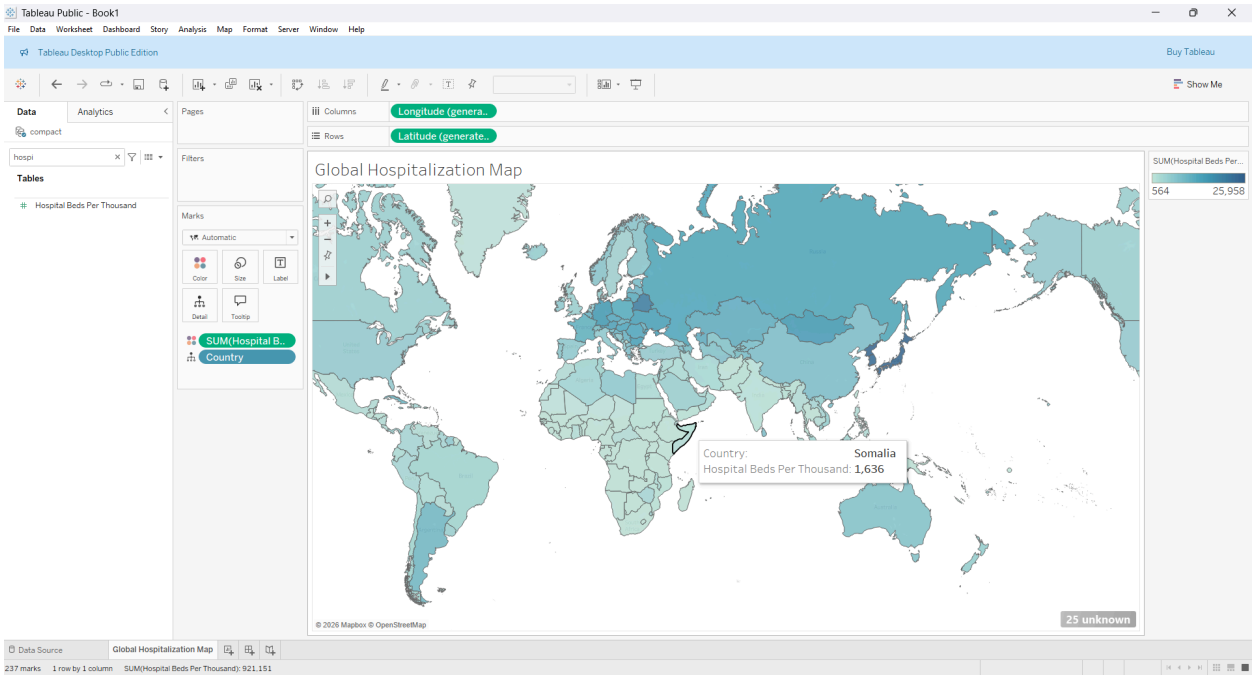


c. COVID-19 Hospitalization and ICU Dashboard
i. Map to show geographic distribution of hospitalization rates
ii. Bar chart to show top 10 countries with highest hospitalization rates
iii. Line chart to show daily new hospitalizations and ICU admissions

Step 1: Create Sheet 1 – Hospitalization Map

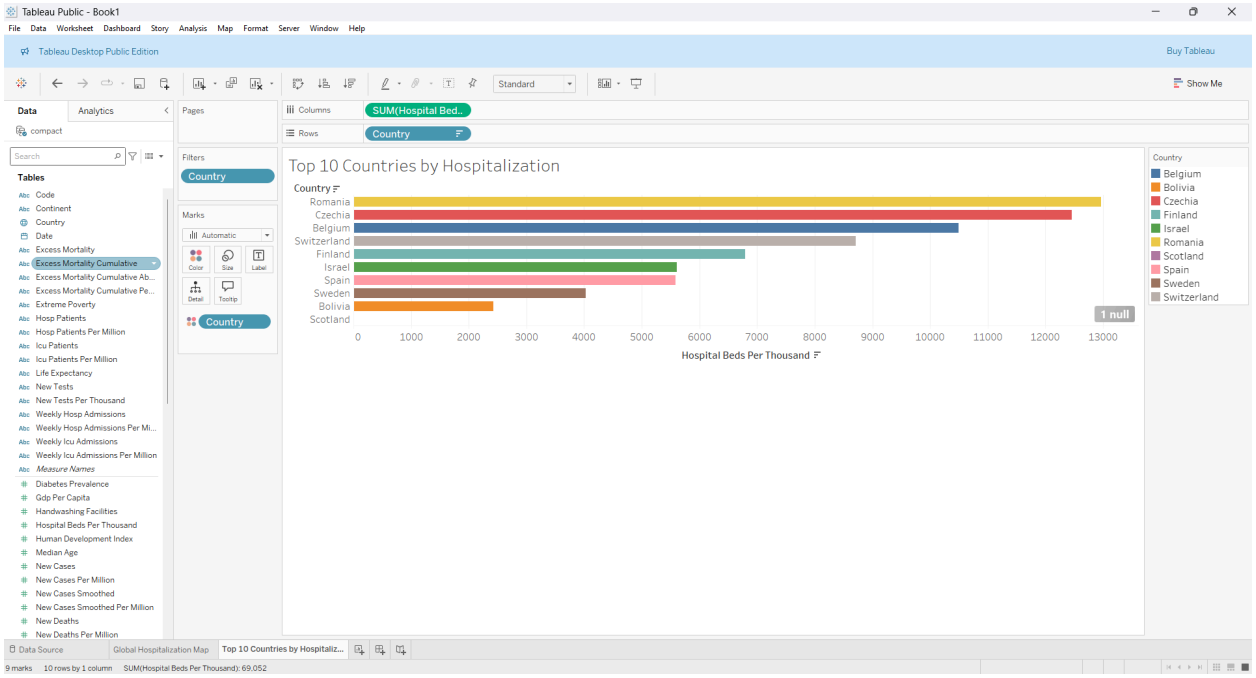
This visualizes which countries had the most strain on their healthcare systems.

1. Click the **New Worksheet** icon.
2. Double-click **Country/Region** to generate the map.
3. Drag **Hospital Beds Per Thousand** (Hospitalized Patients) or **Hospitalization Rate** to the **Color** mark.
4. Change the **Measure** of the pill to **Average** or **Maximum** (Summing hospitalization over time usually results in inflated numbers, as the same person is often counted multiple days).
5. Edit the Colors to a "Sequential" palette like **Blue-Teal** to distinguish it from the "Red" cases map.
6. Rename the sheet: `Global Hospitalization Map`.



Step 2: Create Sheet 2 – Top 10 Countries (Highest Hospitalization)

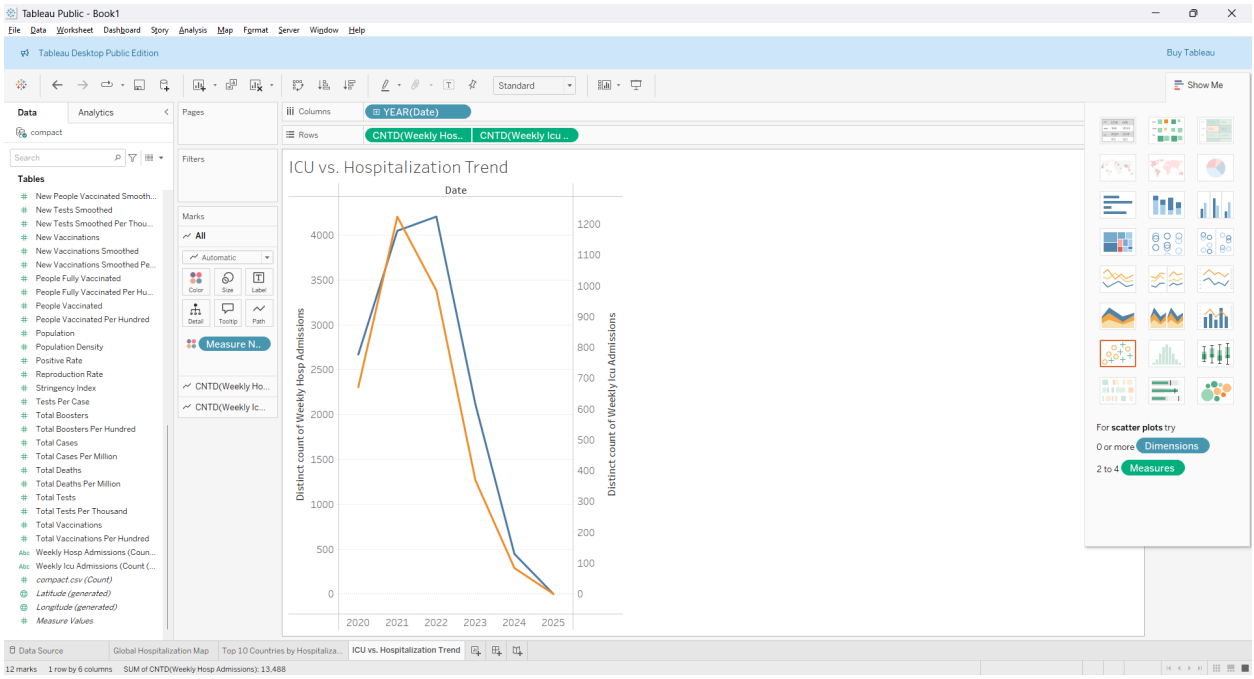
1. Click the **New Worksheet** icon.
2. Drag **Country/Region** to the **Rows** shelf.
3. Drag **Country/Region** to **Color**
4. Drag **Hospital Beds Per Thousand** to the **Columns** shelf.
5. **Top 10 Filter:** * Drag **Country/Region** to the **Filters** shelf.
    - Use the **Top** tab -> **By field** -> **Top 10** -> by **Hosp Patients** (Maximum).
6. Sort descending and add **Labels** to the bars for clarity.
7. Rename the sheet: Top 10 Countries by Hospitalization.



Step 3: Daily/Weekly Hospitalizations & ICU Admissions

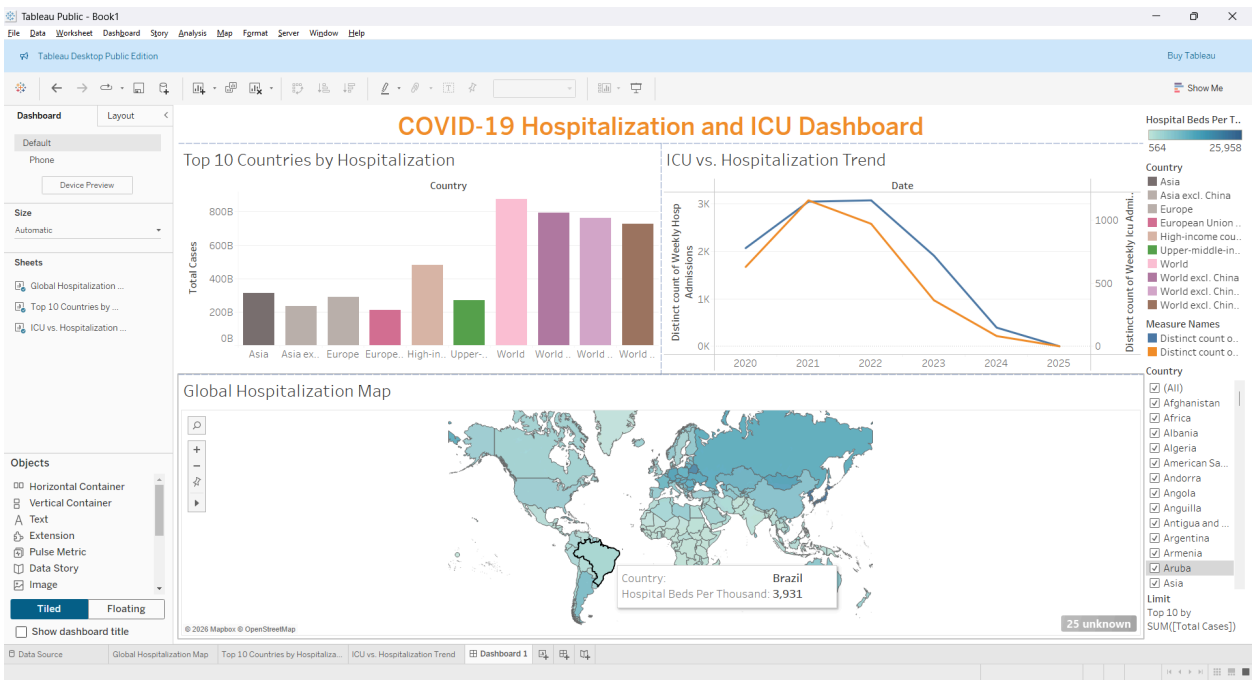
1. Create a New Sheet

- Click the **New Worksheet** icon at the bottom of the screen.
- Double-click the tab and rename it: `Daily Hospitalizations & ICU`.
- Drag **Date** from the Dimensions area to the **Columns** shelf.
- Find **Weekly Hospitalized Patients** in your data list.
- **Important:** If it is under "Dimensions" (blue), drag it down to the **Measures** area (green) first, or simply drag it directly to the **Rows** shelf.
- Now, find **Weekly ICU Patients** and drag it to the **Rows** shelf, placing it right next to the first pill. You will now see two separate graphs, one above the other.
- right-click the **second pill** (Weekly ICU Patients) on the Rows shelf.
- Select **Dual Axis**.



Step 4: Build the Final Dashboard

1. Click **New Dashboard**.
2. **Layout:** * Drag the **Map** to the top half.
    - Place the **Bar Chart** and **Line Chart** side-by-side in the bottom half.
3. **Unified Filters:** * If you already set the Date and Country filters to "Apply to all using this data source" in previous steps, they will appear here automatically.
    - If not, right-click the filter -> **Apply to Worksheets** -> **All using this data source**.
4. **Dashboard Interactivity:**
    - Enable the **"Use as Filter"** (funnel icon) for the Map.
    - Click a country like "Italy" to see how their ICU admissions spiked compared to general hospitalizations.

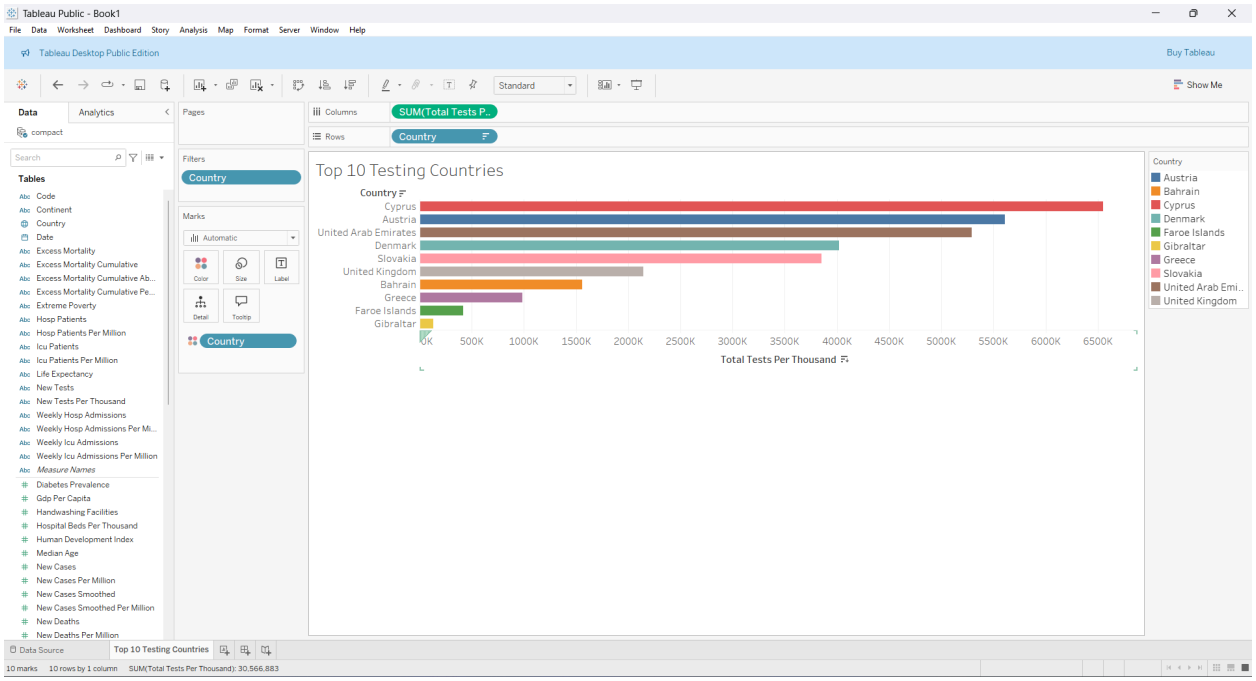


d. COVID-19 Testing and Contact Tracing Dashboard
i. Bar chart to show top 10 countries with highest testing rates
ii. Line chart to show daily new tests and contact tracing efforts
iii. Map to show geographic distribution of testing and contact tracing efforts

Step 1: Create Sheet 1 – Top 10 Testing Rates (Bar Chart)

This identifies which nations were most aggressive in their screening.

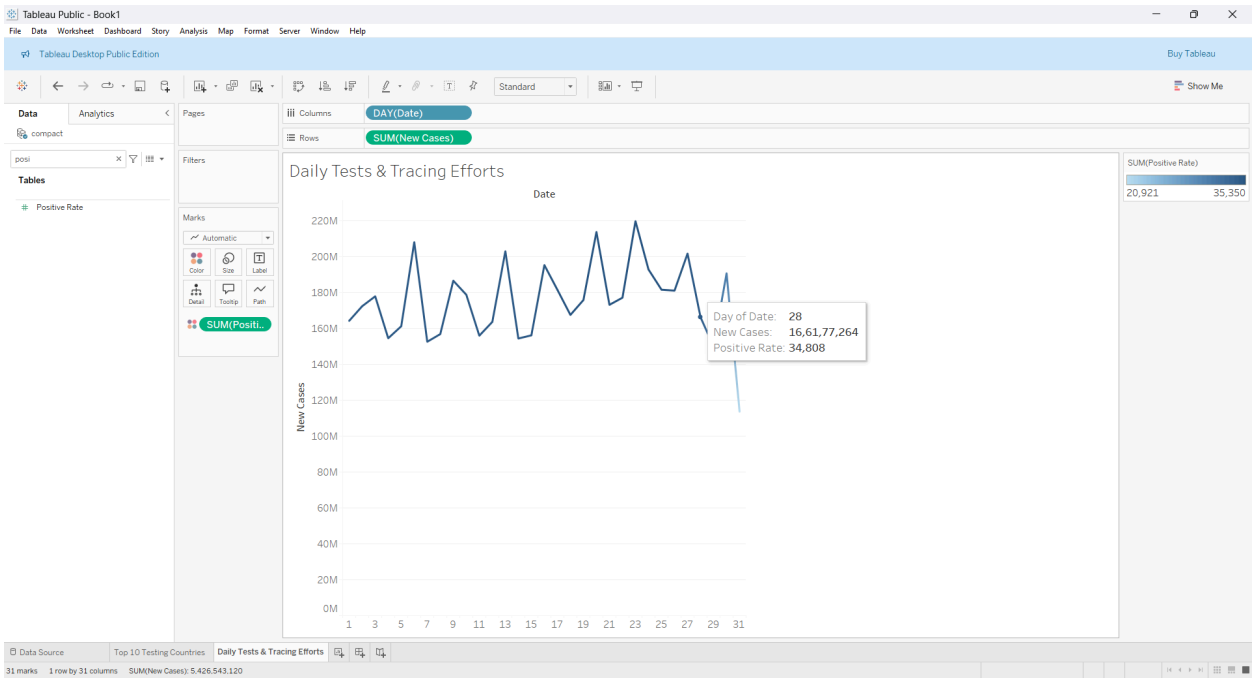
1. Click the **New Worksheet** icon.
2. Drag **Country/Region** to the **Rows** shelf.
3. Drag **Total Tests** (or *Tests per 1,000* if available) to the **Columns** shelf.
4. **Top 10 Filter:**
    - Drag **Country/Region** to the **Filters** shelf.
    - Go to the **Top** tab -> **By field** -> **Top 10** -> by **Total Tests** (Maximum).
5. Sort the bars in descending order.
6. **Color:** Drag **Country/Region** to **Color** to give each bar a unique look as you did in the previous dashboard.
7. Rename the sheet: Top 10 Testing Countries.



Step 2: Create Sheet 2 – Tests and Contact Tracing Trends (Line Chart)

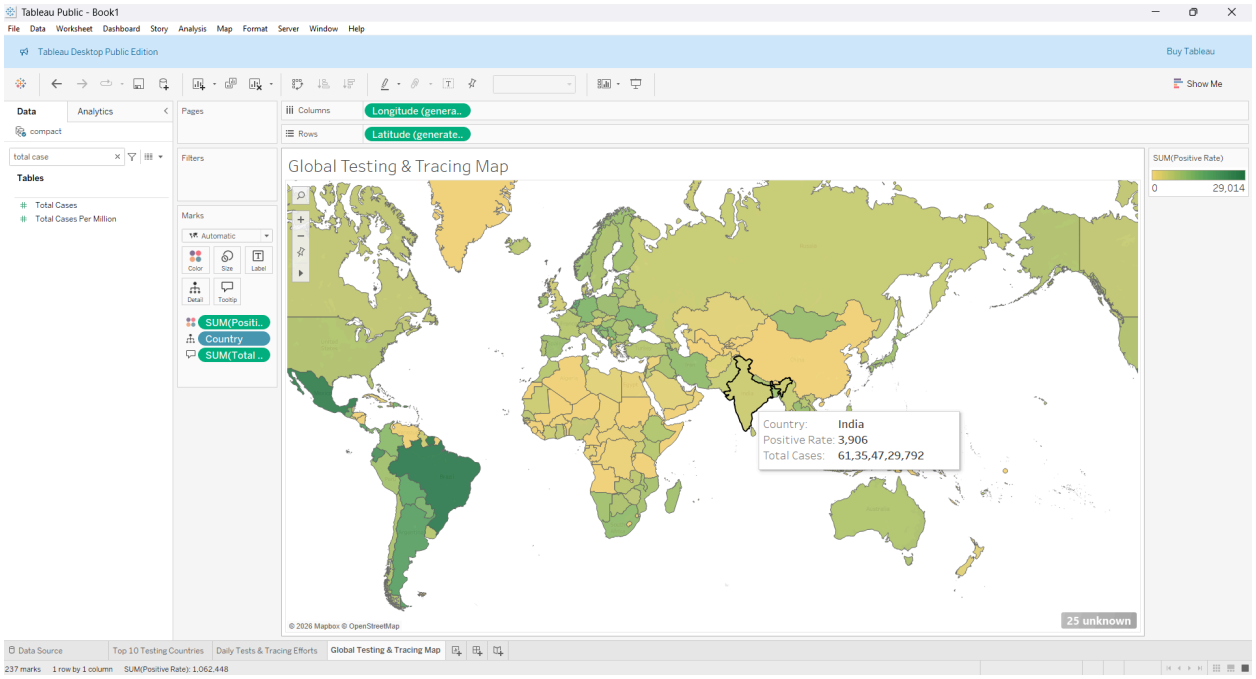
This compares the volume of testing against the policy effort of contact tracing.

1. Click the **New Worksheet** icon.
2. Drag '**date**' to Columns.
3. Drag '**new_tests**' to Rows.
4. Change Marks type to **Line**.
5. Drag '**positive_rate**' to Color to indicate tracing/testing intensity. This shows daily testing activity.
6. Rename the sheet: `Daily Tests & Tracing Efforts`.



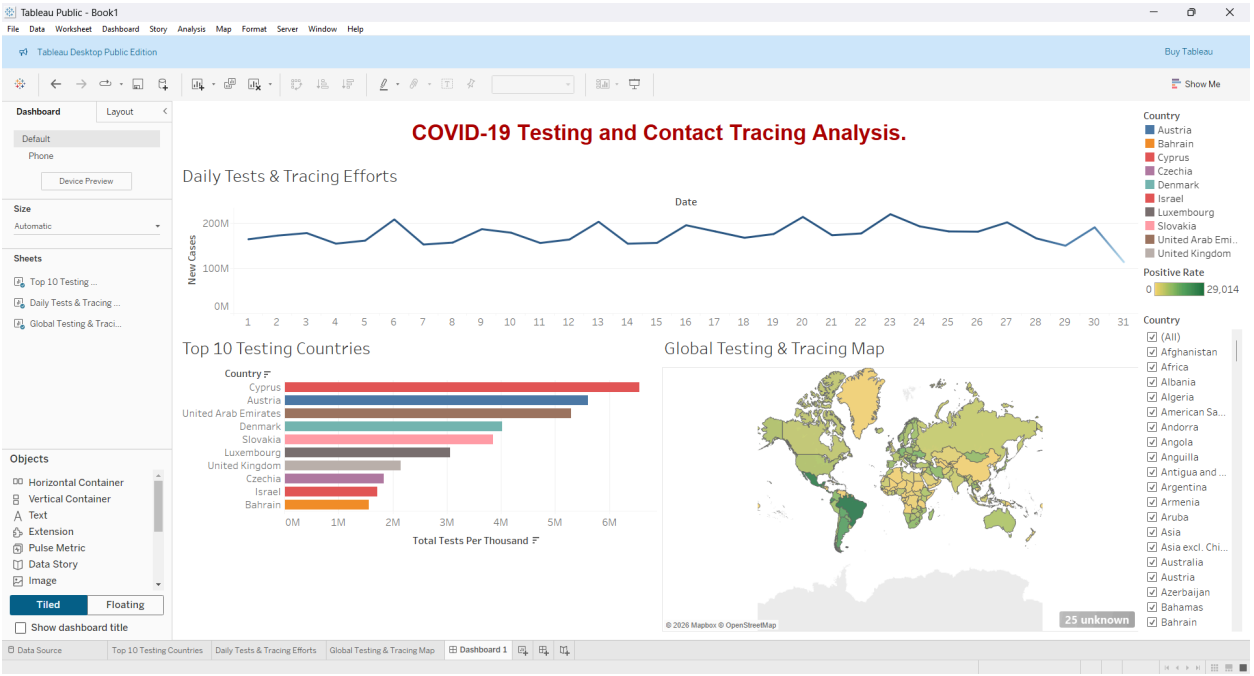
Step 3: Create Sheet 3 – Distribution Map

1. Click the **New Worksheet** icon.
2. Double-click **Country/Region**.
3. Drag **Positive Rate** (Policy Score) to the **Color** mark.
4. Change the color palette to **Green-Gold** or **Viridis** to represent policy intensity.
5. Drag **Total Tests** to the **Tooltip** mark. This way, the color shows the *quality* of tracing, and the size (if used) shows the *volume* of testing.
6. Rename the sheet: `Global Testing & Tracing Map`.



Step 4: Build the Testing & Tracing Dashboard

1. Click **New Dashboard**.
2. **Layout:**
   - Place the **Map** in the center.
   - Place the **Bar Chart** on the left.
   - Place the **Line Trend** chart at the bottom.
3. **Add Interactivity:**
   - Click on the Map and select **Use as Filter** (the funnel icon).
   - Now, when you click on a country, the line chart will show you exactly when that country started its contact tracing program and how their testing scaled.
4. **Filters:** Add the **Date** and **Country/Region** filters to the side. Ensure they are set to **Apply to all using this data source**.
5. **Final Label:** Add a **Text Object** at the top: `COVID-19 Testing and Contact Tracing Analysis`.



1. How to Create a Calculated Field

1. Open any Worksheet in Tableau.
2. In the **Data Pane** on the left, click the small down arrow next to the search bar (or right-click anywhere in the white space of the Data Pane).
3. Select **Create Calculated Field...**
4. A calculation editor window will appear.

ICU to Hospitalization Ratio

This shows what percentage of hospitalized patients require intensive care.

- **Tableau Syntax:** `SUM([ICU Patients]) / SUM([Hosp Patients])`